

Education, Empowerment, and Pathways to Cervical Cancer Risk

I. INTRODUCTION

Cervical cancer is one of the few cancers that is almost entirely preventable, yet it remains a leading cause of female mortality worldwide, with over 662,000 new cases and 348,000 deaths recorded globally in 2022 (1, 2). This burden is unequal: incidence and mortality are two- to five-fold higher in countries where education is lower, reproductive autonomy is constrained, and access to preventive services is limited (1, 2). Cervical cancer is caused primarily by persistent infection with human papillomavirus (HPV), a sexually transmitted virus preventable through vaccination and screening (2). In many low- and middle-income settings, fewer than 5% of women undergo regular cervical screening and HPV vaccination remains low, leaving populations exposed to preventable HPV infection and cervical cancer (2). Despite the availability of preventive methods, inequalities persist as interventions fail to reach socioeconomically disadvantaged and marginalised women, among whom lower knowledge of HPV prevention and reduced uptake of vaccination and screening create gaps that emerge before disease onset and reproduce cervical cancer inequalities (3, 4).

Growing evidence links women's education and empowerment to greater reproductive autonomy, decision-making capacity, and use of preventive health services, while education alone being insufficient without agency and access. (6–10). In the context of cervical cancer, education shapes women's ability to recognise risk and exercise control over reproductive behaviours that determine HPV exposure. Education therefore functions as a foundational determinant of cervical cancer risk rather than a guarantee of protection where empowerment and access remain limited (5). This study builds on this evidence by examining how women's education and empowerment shape reproductive decision-making, behavioural risk, and vulnerability to cervical cancer. By modelling psychosocial and informational pathways linking education and disease risk, this analysis aims to generate actionable insight for targeting cervical cancer prevention among women most at risk (7).

II. ANALYTICAL QUESTIONS & DATA

This study operationalises its aim through three sequential analytical questions mapping the pathway from women's education and empowerment through behavioural risk to cervical cancer vulnerability. Survey-based behavioural data and clinical risk indicators from three UCI Machine Learning Repository datasets are used to translate the conceptual framework into measurable outcomes.

1. *Does women's education and exposure to information predict contraceptive use as a marker of reproductive empowerment?* Using the Contraceptive Method Choice (CMC) (21) dataset, education level and media exposure are modelled as predictors of access to reproductive health information. Contraceptive use (binary) serves as an indicator of reproductive empowerment, capturing preventive reproductive decision-making.
2. *Do empowerment and information access predict behavioural risk relevant to cervical cancer vulnerability?* Using the Cervical Cancer Behavioural Risk (SOBAR) (22) dataset, behavioural sexual risk is modelled as the outcome, predicted by knowledge and awareness-based empowerment measures, with intention and attitude-based variables that capture readiness to act and function as proximal predictors of preventive behaviour (5, 18). Instrumental social support is included as a control to account for structural access constraints on behaviour change (15–16).

3. *Do behavioural risk differences translate into clinical cervical cancer outcomes?*
Using the Cervical Cancer Risk Factors dataset (23), behavioural exposure variables, including STD history and number of sexual partners, are modelled against a clinical cervical outcome variable (biopsy result), linking behavioural risk to clinical vulnerability. Smoking is included as a control due to its independent association with cervical cancer risk (19, 20).

Data Suitability and Limitations

Together, the datasets capture empowerment, behaviour, and disease outcomes. However, the cross-sectional design and self-reported measures limit causal inference and introduce potential reporting bias; findings are therefore interpreted as associative rather than causal.

III. ANALYSIS

Overview of Analytical Strategy

The analysis follows a conceptual pathway linking women's education and empowerment to behavioural risk and, ultimately, cervical cancer vulnerability. Three sequential analytical questions are addressed using regression-based methods applied to complementary datasets from the UCI datasets.

Data Preparation

Question 1: Educational and Informational Predictors of Reproductive Empowerment

Using the CMC dataset, this analysis examined whether women's education and exposure to information predict contraceptive use, operationalised as a behavioural indicator of reproductive empowerment. Predictors were restricted to women's education level and media exposure to isolate informational and educational influences on contraceptive behaviour, avoiding over-adjustment for demographic or household variables that may operate downstream of education in the behavioural pathway.

Contraceptive use was recoded from the original categorical variable into a binary outcome (use vs. non-use) to capture behavioural uptake rather than duration, aligning the outcome with empowerment as engagement in preventive decision-making. Media exposure was recoded so that higher values consistently reflected greater informational access. Missingness checks confirmed complete data for retained variables, removing the need for imputation. Class balance inspection showed a moderate imbalance, supporting stratified train–test splitting and ROC–AUC-based evaluation. Correlation matrices and Variance Inflation Factors (VIF) indicated small-to-moderate associations and low collinearity ($VIF \approx 1$), confirming education and media exposure provided independent information without inflating standard errors or destabilising coefficient estimates.

Question 2: Empowerment, Knowledge, and Behavioural Risk

Using the SOBAR dataset, this analysis examined whether empowerment-related awareness predicts sexual behavioural risk relevant to cervical cancer vulnerability, and whether intention-based psychological mechanisms contribute to how empowerment translates into behaviour beyond structural access constraints.

Predictors were selected to reflect distinct components of the empowerment–behaviour pathway. Empowerment-related knowledge and perceived vulnerability captured awareness and personal risk appraisal, while intention aggregation, commitment, and attitude

consistency represented motivation and follow-through of preventive intentions. These constructs capture whether empowerment is sufficiently internalised to guide behaviour, consistent with evidence on intention-related processes in preventive sexual and reproductive health behaviour (5, 18). Instrumental social support was included as a control to account for material access constraints that may limit behaviour change even when awareness and intention are present. Research demonstrates that practical resources such as affordability, transport, and service accessibility condition women's ability to act on preventive intentions (15, 17). Including this variable allows psychosocial effects to be estimated while accounting for structural barriers. All continuous predictors were standardised to allow comparison of effect sizes across constructs. As the dataset contained no missing values, no imputation procedures were required. Correlation matrices and VIFs confirmed that predictors captured non-redundant information, supporting their simultaneous inclusion and enabling interpretation of independent contributions.

Question 3: Behavioural Risk and Clinical Cervical Cancer Outcomes

Using the Cervical Cancer Risk Factors dataset, this analysis examined whether behavioural sexual risk is associated with a binary clinical cervical outcome, defined as biopsy result (positive vs. negative).

Behavioural risk was captured using two sexual exposure variables: history of sexually transmitted diseases (STDs; any vs. none) and number of sexual partners, representing established pathways to cervical disease via HPV exposure and persistence (16–18). Smoking was included as a control due to its independent carcinogenic role beyond sexual exposure (19, 20). All selected variables were coerced to numeric format to ensure compatibility with logistic regression. Rows with missing values in retained predictors were removed using listwise deletion, reflecting limited missingness and avoiding additional assumptions introduced through imputation. Number of sexual partners was standardised to support interaction modelling and effect size comparability, while binary predictors (STD history and smoking status) were left unscaled. Collinearity diagnostics using correlation matrices and VIF confirmed low intercorrelations, supporting stable coefficient estimation and joint model inclusion.

Analytical Method and Justification

Question 1

Binary logistic regression was selected due to the dichotomous outcome and its ability to estimate odds ratios, enabling direct comparison of education and information exposure within a single behavioural decision context. This modelling choice prioritises interpretability while aligning with the study's analytical questions. An 80/20 train–test split was implemented to assess whether observed associations generalise beyond the training data, with model discrimination evaluated on the held-out test set using ROC–AUC, supporting robustness against sample-specific patterns and motivating progression to Question 2.

Question 2

Hierarchical linear regression was employed to reflect the conceptual ordering of determinants. Empowerment-related awareness and instrumental support were entered in the first block to establish baseline explanatory power. Intention and attitude variables were added in the second block to test whether psychological readiness explains additional variance in behavioural risk beyond awareness and access. This staged approach allows explicit assessment of incremental explanatory contribution. Given the small sample size and the explanatory rather than predictive aim of Question 2, model evaluation prioritised internal

diagnostics, explained variance across model stages, and residual assessment over train–test validation, which would reduce statistical power and limit interpretability.

Question 3

Binary logistic regression was used given the dichotomous biopsy outcome and the need to estimate relative associations between behavioural exposures and cervical pathology. As in Q1, an 80/20 stratified train–test split was used, with model discrimination evaluated on the held-out test set using ROC–AUC to assess generalisation beyond the training data. An interaction between STD history and number of sexual partners was introduced to test whether cumulative sexual exposure amplifies disease risk beyond additive effects, reflecting biological models of HPV persistence (11–13). A sensitivity model excluding smoking was additionally estimated to assess robustness to removal of a major carcinogenic pathway (19, 20), strengthening causal interpretation of behavioural associations.

Model Diagnostics

Question 2

Residual diagnostics were examined for both hierarchical models. Residual–fitted plots indicated broadly linear relationships with mild heteroscedasticity, consistent across blocks. Q–Q plots showed tail deviations from normality, consistent with the skewed and bounded nature of the behavioural risk outcome rather than model misspecification. As these patterns were stable across models and the analysis focuses on estimating relative associations rather than strict distributional inference, linear regression was considered appropriate.

IV. FINDINGS, REFLECTIONS AND FURTHER WORK

Findings

A. Does women's education and exposure to information predict contraceptive use as marker of reproductive empowerment?

Education and reproductive health information were positively associated with contraceptive use. Higher educational attainment was associated with increased contraceptive use (OR = 1.64, 95% CI [1.44, 1.86], $p < .001$), with predicted probabilities increasing monotonically across education levels (Figure 1). Media exposure independently strengthened this association (OR = 1.71, 95% CI [1.04, 2.81], $p = .034$), indicating that informational access beyond formal schooling adds explanatory value. Model discrimination was modest but above chance (ROC–AUC ≈ 0.60).

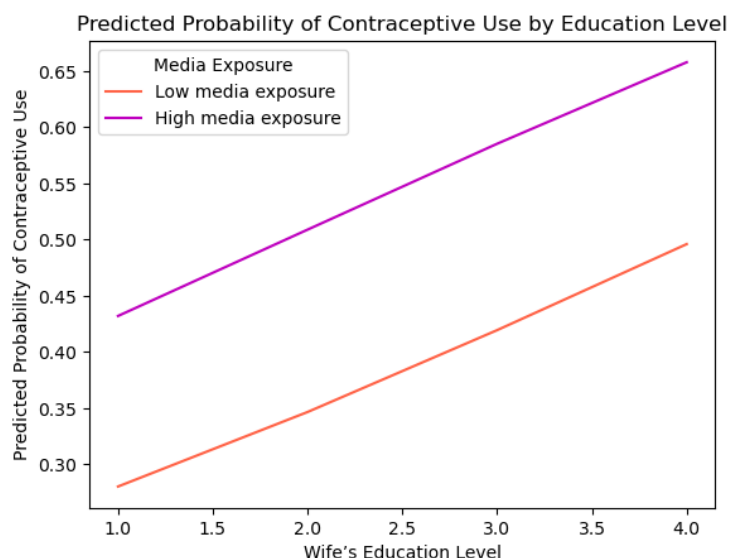


Figure 1. Predicted probability of contraceptive use by women's education level, stratified by media exposure (Question 1).

B. Do empowerment and information access predict behavioural risk relevant to cervical cancer vulnerability?

In contrast, empowerment-related factors showed limited explanatory power for sexual behavioural risk. Measures of perceived vulnerability, intention strength, and attitude consistency explained little variance in behavioural risk ($\Delta R^2 \approx 0.03$; Figure 2), indicating that awareness and motivational readiness alone do not reliably translate into reduced sexual risk.

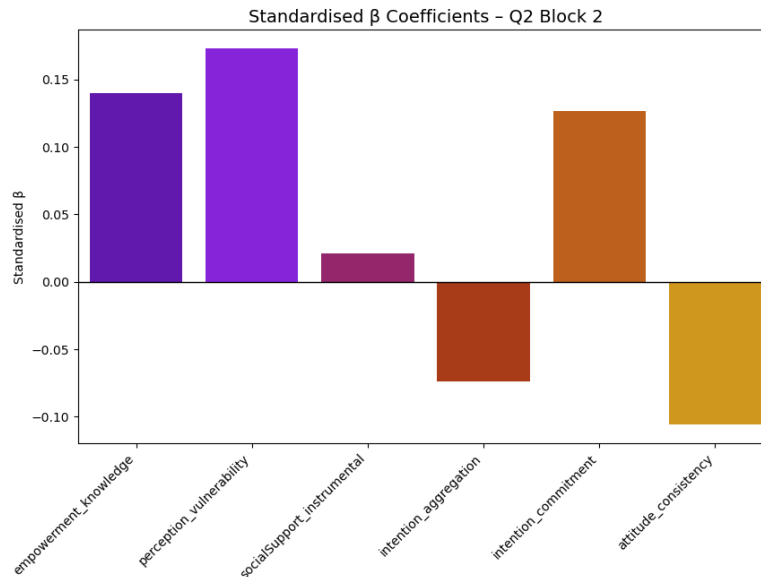


Figure 2. Standardised coefficients from hierarchical regression models predicting sexual behavioural risk (Question 2).

C. Do behavioural risk differences translate into clinical cervical cancer outcomes?

Behavioural sexual exposure demonstrated clearer clinical relevance. History of STDs approximately doubled the odds of biopsy-confirmed cervical pathology across model specifications (OR range ≈ 2.0 – 2.1 ; Figure 3), with the association reaching statistical significance in sensitivity analyses excluding smoking (OR = 2.06, 95% CI [1.01, 4.68], $p = .047$). Number of sexual partners did not independently predict biopsy outcome after accounting for STD history (OR ≈ 0.99 , 95% CI [0.69, 1.41], $p > .05$; Figure 4), indicating that partner count is a coarse risk proxy.

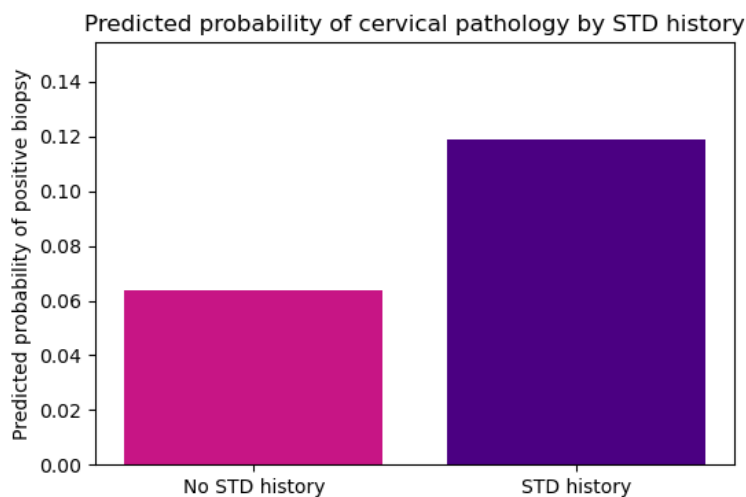


Figure 3. Predicted probability of biopsy-confirmed cervical pathology by STD history, estimated from logistic regression marginal effects (Question 3).

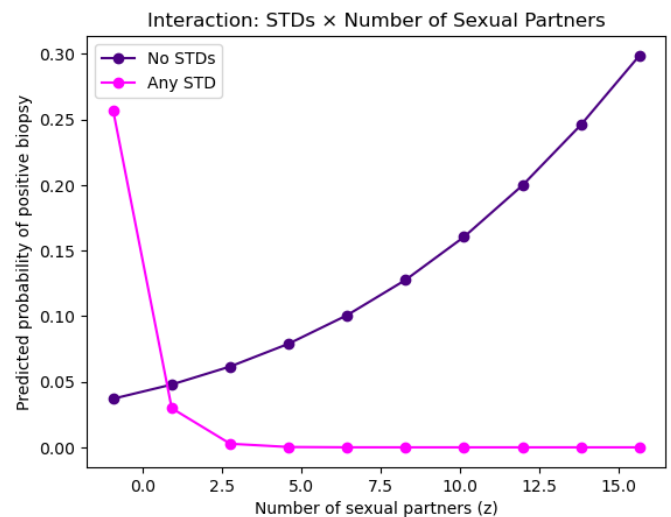


Figure 4. Interaction between STD history and number of sexual partners predicting biopsy-confirmed cervical pathology (Question 3).

vulnerability. Although cross-sectional design limits causal inference, the staged approach allowed each segment of the pathway to be evaluated appropriately.

Q1 findings indicate that education and media exposure empower preventive reproductive decision-making, with modest model discrimination ($AUC \approx 0.60$) indicating substantial unexplained variation. This aligns with evidence that health messaging effectiveness depends on quality and context (21) of messaging, and that informal or community-based education can outperform formal schooling (22). Future work should refine educational and informational measures by accounting for message quality and context.

In Q2, empowerment-related knowledge, perceived vulnerability, and intention measures explained little variance in sexual behavioural risk. This aligns with the well-documented intention–behaviour gap in sexual health, where awareness does not reliably translate into safer behaviour (26). Prior research highlights relational power, partner dynamics, and structural constraints in shaping sexual risk (23–25), none of which are captured here. This limited explanatory power likely reflects construct measurement limitations rather than the modelling approach.

Q3 showed the strongest link to health outcomes, as STD history was consistently associated with biopsy positivity across model specifications, with the association most clearly supported in sensitivity analyses, consistent with established cervical cancer aetiology (11–14). Number of sexual partners does not independently predict biopsy outcome after accounting for STD history, indicating that partner count is a coarse proxy for cervical cancer risk. Consistent with prior evidence, documented exposure captures clinically relevant sexual risk more directly than behavioural frequency alone (27–29).

Overall, education and information access support empowered reproductive decision-making, but do not reliably translate into reduced cervical cancer vulnerability. Sexual risk and clinical outcomes are shaped predominantly by cumulative behavioural exposure than by knowledge or intention, underscoring a gap between empowerment and health outcomes. Generalisability is strongest for reproductive decision-making (Q1), weaker for sexual behavioural risk (Q2), and moderate for clinical outcomes (Q3). Future work should prioritise

longitudinal datasets capturing education, empowerment, behavioural trajectories, and cervical outcomes within the same individuals, enabling clearer identification of why empowerment fails to translate into health protection.

Word Count

- Introduction (296/300)
- Analytical questions and data (277/300)
- Analysis (982/1000)
- Findings, reflections and further work (599/600)

AI Appendix

1. Data Preparation and Variable Labelling (Question 1)

- **Purpose of AI use:** To assist with improving the readability and interpretability of raw datasets by assigning meaningful column names based on accompanying documentation.
- **Location in assignment:** Data preparation and preprocessing (Question 1).
- **AI tool:** ChatGPT.
- **Prompt:** “I am working with a UCI dataset that does not contain column headers. The documentation provides the variable order. How can I assign column names when loading the data in Python to make the dataset more readable?”
- **AI response:** Suggested supplying a list of column names directly to `pd.read_csv()` using the `names=` argument, based on the dataset documentation.
- **Modification / implementation:** Column names were manually verified against the official `cmc.names` file before implementation to ensure accuracy. The approach was then applied consistently across files where headers were absent.

2. Odds Ratios and Confidence Intervals for Logistic Regression (Question 1)

- **Purpose of AI use:** To clarify how to compute and present odds ratios and 95% confidence intervals from a fitted logistic regression model.
- **Location in assignment:** Statistical modelling and results reporting (Question 1).
- **AI tool:** ChatGPT.
- **Prompt:** “I have fitted a logistic regression model using `statsmodels`. How do I calculate odds ratios and 95% confidence intervals from the model parameters?”
- **AI response:** Explained how to extract model coefficients using `.params`, confidence intervals using `.conf_int()`, and exponentiate these values to obtain odds ratios and confidence intervals.

3. Predicted Probabilities and Stratified Visualisation (Question 1)

- **Purpose of AI use:** To assist with generating predicted probabilities of contraceptive use across education levels, stratified by media exposure.
- **Location in assignment:** Visual analytics and model interpretation (Question 1).
- **AI tool:** ChatGPT.
- **Prompt:** “How can I generate predicted probabilities from a logistic regression model across education levels, with separate lines for high and low media exposure?”
- **AI response:** Suggested constructing a prediction grid by duplicating education levels and varying the media exposure variable, then passing this grid to the model’s `predict()` method.
- **Modification / implementation:** The prediction dataframe was aligned to the model design matrix using the training columns to avoid dimension mismatch errors.

4. Standardised Coefficients and Model Interpretation (Question 2)

- **Purpose of AI use:** To assist with extracting and inspecting regression coefficients from a fitted standardised model to support interpretation of relative predictor importance.
- **Location in assignment:** Hierarchical regression and coefficient interpretation (Question 2)
- **AI tool:** ChatGPT.
- **Prompt:** “I have fitted a regression model using standardised variables. How can I extract the coefficients to compare the relative contribution of each predictor?”
- **AI response:** Explained how to access model coefficients using the `.params` attribute of the fitted `statsmodels` object.

5. Plot Refinement and Visual Clarity (Questions 2 and 3)

- **Purpose of AI use:** To improve interpretability of plots by addressing visual issues such as overlapping labels and reference lines.
- **Location in assignment:** Figures and visual presentation (Questions 2 and 3).
- **AI tool:** ChatGPT.
- **Prompts:**
 - “How can I add a horizontal reference line at zero to this coefficient plot?”
 - “My x-axis labels are overlapping, how can I fix this?”
- **AI response:** Suggested adding a horizontal reference line using `plt.axhline(0)` and rotating tick labels with adjusted alignment.
- **Modification / implementation:** These adjustments were applied selectively where they improved readability without altering the underlying results.

6. Interaction Effects and Predicted Probabilities (Question 3)

- **Purpose of AI use:** To assist with constructing a prediction grid and visualising the interaction between STD history and number of sexual partners on predicted biopsy probability from a fitted logistic regression model.
- **Location in assignment:** Interaction modelling and visualisation of marginal effects (Question 3).
- **AI tool:** ChatGPT.
- **Prompt:** “I have fitted a logistic regression model with an interaction between STD history and number of sexual partners. How can I generate predicted probabilities across the range of partners and plot separate lines for $STD = 0$ and $STD = 1$?”
- **AI response:** Suggested creating a grid of values spanning the observed range of the continuous predictor, computing the interaction term manually, passing the grid to the fitted model to obtain predicted probabilities, and plotting results stratified by the binary variable.
- **Modification / implementation:** Smoking was fixed at zero to isolate the interaction of interest. The prediction grid was manually constructed to match the model design matrix.

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